RUBIK'S CUBESOLVE



Under the Esteemed Guidance of

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AIM

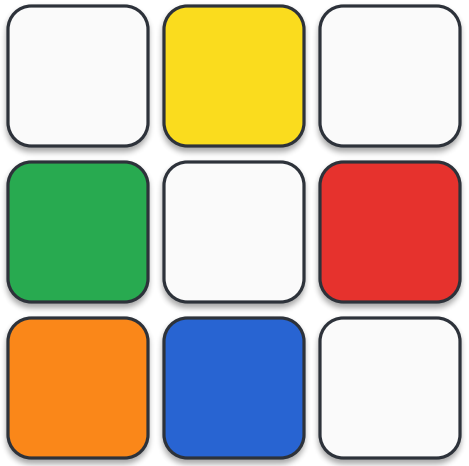
To develop a web-based Rubik's Cube Solver using React.js that scans the cube through a camera, detects its colours, validates the cube and generates step-by-step solving instructions.

The system helps beginners understand and follow the solving process easily.

PROBLEM STATEMENT

Solving a Rubik's Cube manually requires practice, patience and logical thinking. Beginners often find it difficult to identify the correct sequence of moves and understand how each move changes the cube.

- Manual solving can be confusing for beginners.
- Identifying all six face colours correctly is difficult.
- A clear, guided solution is needed to reduce errors.



A cube has 6 faces and 54 coloured stickers.

PROBLEM EXPLANATION

The application guides a user from cube scanning to a verified solution.

01 **Scan the cube**

Capture or upload images of all six faces.

02 **Detect colours**

Identify the nine sticker colours on each face.

03 **Validate cube**

Check sticker count, centres and piece arrangement.

04 **Generate moves**

Calculate a valid solving sequence and show it step by step.

IMPLEMENTATION

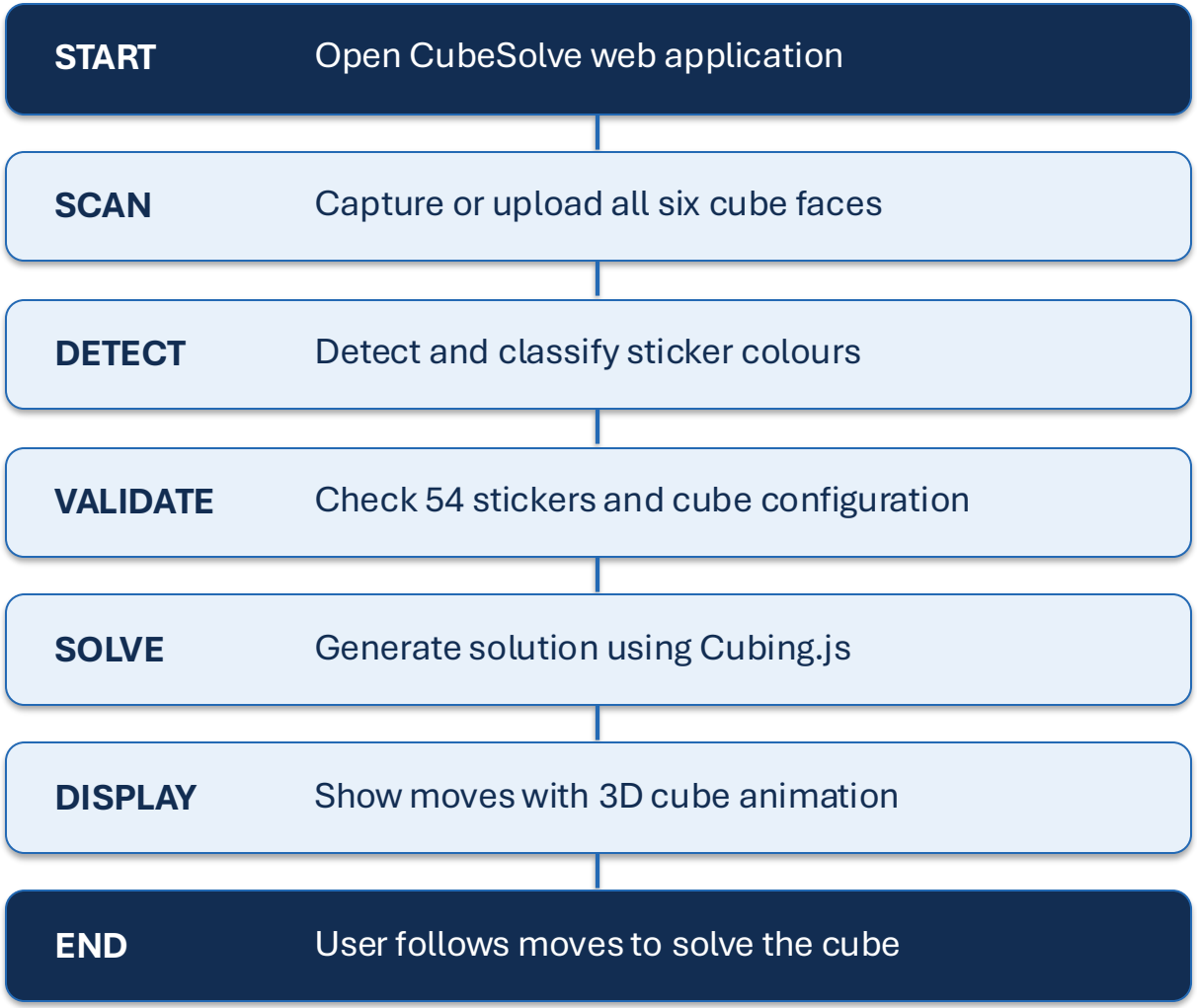
The project is implemented as a React-based web application.

Frontend	React.js, JavaScript, HTML, CSS and Tailwind CSS
Development Tool	Vite for running and building the application
Cube Logic	Cubing.js to calculate the solution and moves
Colour Processing	CIE Lab colour comparison using colour distance (deltaE)

Main Features

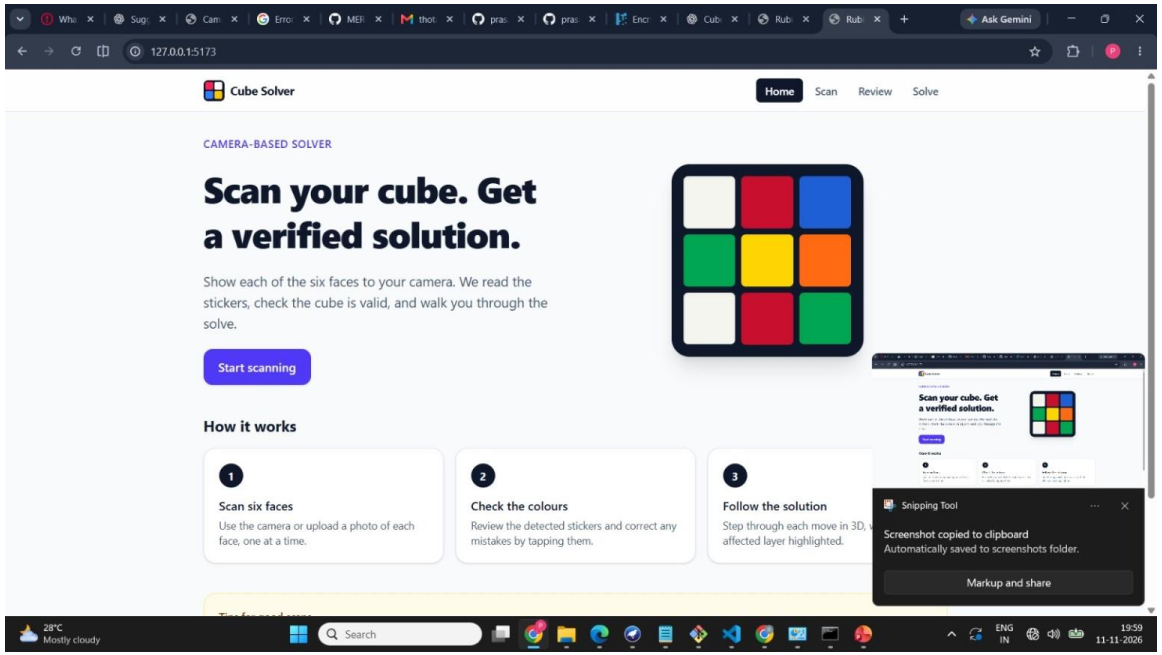
- Camera scan / image upload
- Six-face colour detection
- Cube validation
- Step-by-step move display
- Next, Previous, Reset and Auto-play

FLOW CHART


START
Open CubeSolve web application
SCAN
Capture or upload all six cube faces
DETECT
Detect and classify sticker colours
VALIDATE
Check 54 stickers and cube configuration
SOLVE
Generate solution using Cubing.js
DISPLAY
Show moves with 3D cube animation
END
User follows moves to solve the cube

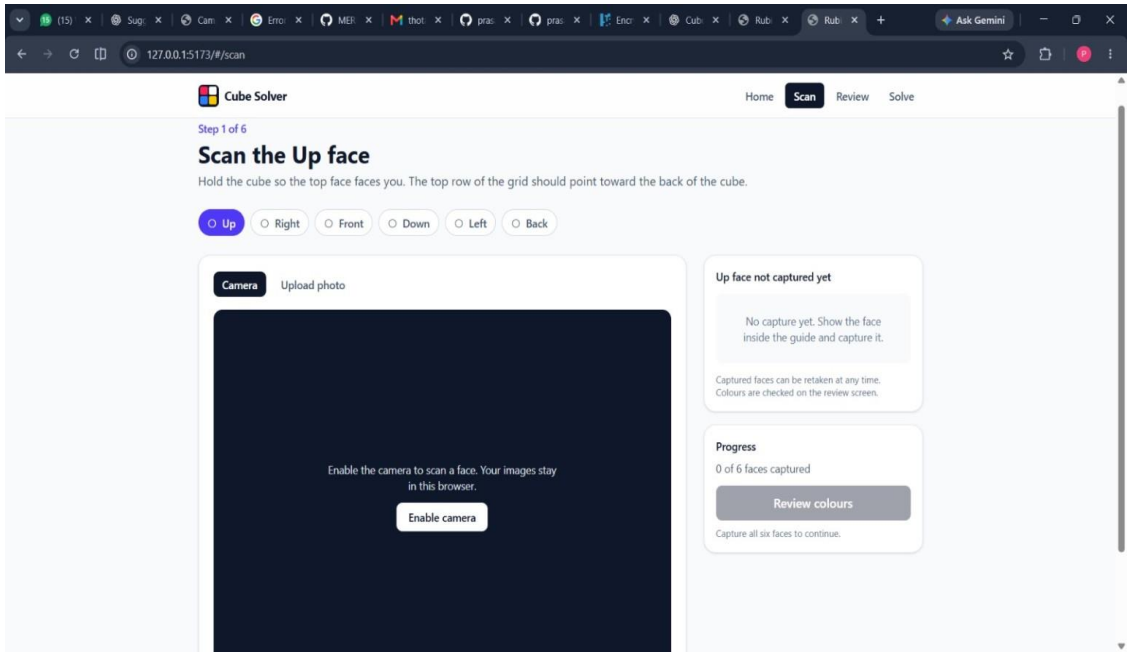
OUTPUT

Screenshots of the implemented CubeSolve web application



Home Page

Camera-based cube solver introduction and scanning entry point.



Scan Page

Six-face capture interface with camera and image-upload options.

CONCLUSION

Rubik's CubeSolve provides a simple and guided way for beginners to solve a Rubik's Cube.

- Scans or accepts images of all six cube faces.
- Detects colours and checks whether the cube configuration is valid.
- Uses Cubing.js to generate a sequence of solving moves.
- Displays step-by-step instructions with controls such as Next, Previous, Reset and Auto-play.

THANK YOU

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